Proposition 13: no linear bound for the spectral distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Proposition13_SpectralLowerBound` formalises Proposition 13 with the construction of Appendix F.1, Lemma 49, Theorem 3 (4) and the remark after Proposition 13.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.Prop13.proposition13`: for every $r \geq 1$, with $n = |V(G_r)| = 4r^2 + 3r + 1$,

$$d_2(G_r, H_r) = \sqrt{r} \qquad \text{and} \qquad \tfrac{1}{2\sqrt{2}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1;$$

`GsimOT.Prop13.proposition13_explicit` adds $\frac{1}{2\sqrt{2}} \leq \frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$ (Lemma 49). `GsimOT.Prop13.no_power_bound`: for every $\alpha > \frac{3}{4}$ and C, some G_r, H_r violate $\delta_2 \leq C(\delta_2^{\mathrm{OT}})^{\alpha}$.

Correspondence

Paper	Page	Statement	Lean declaration(s) and location
Proposition 13	11	$d_2(G_r, H_r) = \sqrt{r}$ and $\frac{1}{2\sqrt{2}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1$, $n = 4r^2 + 3r + 1 = \Theta(r^2)$	`GsimOT.Prop13.proposition13` `Proposition13_SpectralLowerBound/Proposition13.lean:46`
Prop. 13 + Lemma 49	11, 45	the same with $\frac{1}{2\sqrt{2}} \leq \frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$, the bound $\frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$ being Lemma 49	`GsimOT.Prop13.proposition13_explicit` `Proposition13_SpectralLowerBound/Proposition13.lean:25`
Theorem 3 (4)	4	$\delta_2(G_r, H_r) \geq \sqrt{r} \cdot \delta_2^{\mathrm{OT}}(G_r, H_r)$	`GsimOT.Prop13.theorem3_item4` `Proposition13_SpectralLowerBound/Proposition13.lean:81`

after Prop. 13	11	No bound $\delta_2 \leq C_\alpha(\delta_2^{\mathrm{OT}})^{\alpha}$ for $\alpha > \frac{3}{4}$ (l. 391–393)	`GsimOT.Prop13.no_power_bound`
			`Proposition13_SpectralLowerBound/Proposition13.lean:91`
			`GsimOT.Prop13.deltaPQ_eq`
			`Proposition13_SpectralLowerBound/Proposition13.lean:60`
			`GsimOT.Prop13.deltaOTPQ_bounds`
			`Proposition13_SpectralLowerBound/Proposition13.lean:67`

Paper	Page	Statement	Lean declaration(s) and location
Appendix F.1, Fig. 7	45	Core $X = \{u\} \cup \{v_{i,k}\}$, clocks C, tags T_i; $N(v_{i,k}) = T_i \cup \{c_1, \ldots, c_k\}$; u isolated in G_r, adjacent to C in H_r	GsimOT.Prop13.Core Proposition13_SpectralLowerBound/Construction.lean:27 GsimOT.Prop13.Marker Proposition13_SpectralLowerBound/Construction.lean:30 GsimOT.Prop13.Vtx Proposition13_SpectralLowerBound/Construction.lean:33 GsimOT.Prop13.graph Proposition13_SpectralLowerBound/Construction.lean:61 GsimOT.Prop13.Gr Proposition13_SpectralLowerBound/Construction.lean:67 GsimOT.Prop13.Hr Proposition13_SpectralLowerBound/Construction.lean:70
Appendix F.1	45	$\lvert V\rvert = n = 4r^2 + 3r + 1$; H_r has minimum degree $\geq r$	GsimOT.Prop13.card_Vtx Proposition13_SpectralLowerBound/Construction.lean:35 GsimOT.Prop13.le_deg_Hr Proposition13_SpectralLowerBound/Construction.lean:138
Appendix F.1	45	$d_2(G_r, H_r) = \sqrt{r}$: row of $\pi(u)$ (spectral norm $\geq$ row norm) and the star for id (l. 1412–1416)	GsimOT.Prop13.distPQ_eq Proposition13_SpectralLowerBound/SpectralDistance.lean:120 GsimOT.Prop13.sqrt_row_le_opNorm22 Proposition13_SpectralLowerBound/SpectralDistance.lean:47 GsimOT.Prop13.sqrt_le_opNorm_alignedDiff Proposition13_SpectralLowerBound/SpectralDistance.lean:73 GsimOT.Prop13.opNorm_diff_refl_le Proposition13_SpectralLowerBound/SpectralDistance.lean:101
Lemma 49	45	$\frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1$	GsimOT.Prop13.lemma49 Proposition13_SpectralLowerBound/Lemma49.lean:420
Proof of Lemma 49	45–46	Shifts σ_i and $Q = \frac{1}{2r} \sum_i P_{\sigma_i}$	GsimOT.Prop13.shift Proposition13_SpectralLowerBound/Lemma49.lean:113 GsimOT.Prop13.tau Proposition13_SpectralLowerBound/Lemma49.lean:129 GsimOT.Prop13.Qs Proposition13_SpectralLowerBound/Lemma49.lean:137 GsimOT.Prop13.Qs_isDS Proposition13_SpectralLowerBound/Lemma49.lean:170

Paper	Page	Statement	Lean declaration(s) and location
Proof of Lemma 49	46	Vectors d_{xy}; $C_2(Q) \le 1$ (tags: Cauchy–Schwarz, clocks orthogonal; l. 1424–1436)	GsimOT.Prop13.dv Proposition13_SpectralLowerBound/Lemma49.lean:184 GsimOT.Prop13.sum_qc_dvDot_le Proposition13_SpectralLowerBound/Lemma49.lean:239 GsimOT.Prop13.opNorm_Bq_le Proposition13_SpectralLowerBound/Lemma49.lean:386 GsimOT.Prop13.dOT_le_one Proposition13_SpectralLowerBound/Lemma49.lean:400
Proof of Lemma 49	46	Test vector $\mathbf{1}_R$: $(B\mathbf{1}_R)(x,y) = \mathbf{1}_R(x,y)(\deg_G x - \deg_H y)$, hence $C_2(R) \ge r/\sqrt{n}$ (l. 1437–1446)	GsimOT.Prop13.le_opNorm_Bmat Proposition13_SpectralLowerBound/Lemma49.lean:50 GsimOT.Prop13.le_dOT_of_deg Proposition13_SpectralLowerBound/Lemma49.lean:101 GsimOT.Prop13.le_dOT Proposition13_SpectralLowerBound/Lemma49.lean:407
Proof of Prop. 13	46	$\frac{r}{\sqrt{n}} \ge \frac{1}{2\sqrt{2}}$ since $8r^2 \ge n$ (l. 1447–1448)	GsimOT.Prop13.proposition13_explicit Proposition13_SpectralLowerBound/Proposition13.lean:25

Shared notions used (from GsimOT/Common)

Paper	Page	Statement	Lean declaration(s) and location
Section 1, Appendix A	2–3, 14	Couplings $\mathrm{DS}^*(V,W)$; x^a with $0^a := 0$ (convention $0^0 = 0$)	GsimOT.IsCoupling Common/OTDistances.lean:36 GsimOT.powZ Common/OTDistances.lean:42
Section 1, Eqs. (1), (2)	2–3	$B(G,H,Q)$ and $B_{p\to q}(G,H,Q)$	GsimOT.B Common/OTDistances.lean:58 GsimOT.Bmat Common/OTDistances.lean:70

Paper	Page	Statement	Lean declaration(s) and location
Eq. (2), Section 2	3, 5	$\delta^{\mathrm{OT}}_{p\to q}$ (couplings) and $d^{\mathrm{OT}}_{p\to q}$ ($\mathrm{DS}(V,W)$), both infima	`GsimOT.deltaOTPQ` `Common/OTDistances.lean:74` `GsimOT.dOT` `Common/OTDistances.lean:79`
Sections 1–2, Appendices A–B	3, 5, 14–15	$\delta^{\mathrm{OT}}_{p\to q} = n^{-\kappa_{p\to q}} d^{\mathrm{OT}}_{p\to q}$ for graphs of the same order n ($d^{\mathrm{OT}}_{p\to q}$ is the unnormalised version, $s(n) = n^{\kappa_{p\to q}}$), via $\mathrm{DS}(V,W) = n\,\mathrm{DS}^*(V,W)$ and the homogeneity of $B_{p\to q}$ and of $\|\cdot\|_{p\to q}$	`GsimOT.deltaOTPQ_eq_dOT` `Common/OTDistances.lean:131` `GsimOT.Bmat_smul` `Common/OTDistances.lean:117` `GsimOT.opNorm_smul_of_nonneg` `Common/OTDistances.lean:94`
Appendix A	14–15	"All minima … are attained" for $\delta^{\mathrm{OT}}_{p\to q}$ and $d^{\mathrm{OT}}_{p\to q}$: lower semicontinuity of $Q \mapsto \|B_{p\to q}(G,H,Q)\|$, compactness	`GsimOT.lowerSemicontinuous_opNorm_Bmat` `Common/Attainment.lean:136` `GsimOT.isCompact_setOf_isCoupling` `Common/Attainment.lean:168` `GsimOT.exists_deltaOTPQ_eq` `Common/Attainment.lean:199` `GsimOT.exists_dOT_eq` `Common/Attainment.lean:223`

Notes on the formalisation

1. **Upper bound of Lemma 49.** The paper computes $C_2(Q)^2 = \|K^\top K\|_2 = 1$ for the block form $\begin{pmatrix} 0 & K \\ K^\top & 0 \end{pmatrix}$. The formalisation instead shows $\sum_{(x,y)} Q(x,y)(d_{xy}\cdot\zeta)^2 \le \|\zeta\|^2$. It uses Cauchy–Schwarz on each tag block and the orthogonality of the clock markers, and concludes $|y^\top B x| \le 1$ through Lemma 19. This is the same estimate, organised as a bilinear bound.

2. The lower bound of Lemma 49 is proved for arbitrary graphs (`le_dOT_of_deg`: a vertex whose degree differs from every degree in H by $\ge c$ forces $d_2^{\mathrm{OT}} \ge c/\sqrt{n}$) and then specialised to $c = r$.

3. The remark after Proposition 13 is formalised as `no_power_bound`. It uses $\delta_2(G_r,H_r) = \sqrt{r}/n$, $\delta_2^{\mathrm{OT}}(G_r,H_r) \le 1/n$ and $n \le 8r^2$; the lower estimate $\delta_2^{\mathrm{OT}} = \Omega(n^{-1})$ is not needed for this conclusion.

4. Indices: $v_{i,k}$ is `some (i, k)` and u is `none` in `Core`; clocks and tags are the two summands of `Marker`. G_r and H_r are one construction `graph` with a Boolean flag for the edges at u.

5. `proposition13` is Proposition 13. `proposition13_explicit` states Proposition 13 together with the bound $\frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$ of Lemma 49, which the proof of Proposition 13 combines with $\frac{r}{\sqrt{n}} \geq \frac{1}{2\sqrt{2}}$; `proposition13` is derived from it.